



Geomec 5.1 Tutorial

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Table 1: Document Change Record

Date	Version	Author	Comments
March 6, 2014	V0.1 Draft	Alfred van der Hoeven	First version
July 29, 2014	V0.2 Draft	Alfred van der Hoeven	Second version
October 28, 2014	V0.3 Draft	Alfred van der Hoeven	5.0.0 release version
January 2015	V0.4	Alfred van der Hoeven	5.1.0 release version



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0 Executive Summary

This document describes how the Open Inventor version of Geomec has changed from the OpenGL version of Geomec and how to apply these changes to best effect.

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1 Introduction

To speed up and improve the visualization of the elements in a Geomec model, a new version of Geomec has been developed that employs Open Inventor and Open Inventor MeshViz XLM for the graphical representation of the Geomec model.

This document describes the changes from the OpenGL version of Geomec to the Open Inventor version of Geomec in the form of a tutorial.

1.1 Key Assumptions

- This document is aimed at people with a fair amount of knowledge of the Geomec application.

2 Overview of what has changed

2.1 Views

Views that have disappeared:

- Birdeye view
- Cross section view
- Default view
- Top view

View that has replaced the above:

- Main view (OIV)

The following 2D views have been replaced as well:

- History View
- Wellpath Results View

2.2 New toolbar buttons



The *coordinate axis* button will show/hide a set of coordinate axes.



The *element/node* button allows you to switch between a node- and element-based visualization mode.



The *mesh* button allows you to toggle the display of mesh edges.



The *legend* button allows you to toggle the display of a legend



The *go to saved* position button will reset the camera position to a previously saved position.



The *save position* button will save the current camera position



The *formation outlines* button allows you to visualize formation outlines on a cross section.



The *contour line* button allows you to visualize contour lines and to select a property for which to visualize contour lines.

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The *clipping* button allows you to turn clipping on/off when displaying cross sections



The *dragger-based value tracker* button allows you to visualize or hide the 3D value dragger and the mesh properties display window



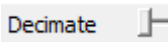
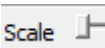
The *mouse-based value tracker* button will display the mesh properties display window and lets you select mesh elements with the mouse left button.



The *tensor vector* buttons can be used to display the full, normal or shear component of the vector result of a tensor as it is applied to a cross section



The *beach ball* button allows you to toggle the display of tensor beach balls.



The *scale* and *decimate* sliders allow you to manipulate the size and number of vectors or beach balls displayed when displaying vectors or beach balls.

2.3 Modified dialogs

2.4 Legend view

The legend has moved from the separate legend view window into the main window. You are advised to close this separate legend view window.

3 A guided tour of the new Open Inventor view

3.1 Initial impression

After opening the application and after opening a model by creating a new one or by reading in a model file, a view will appear, this is the main (OIV) view.

As in the OpenGL Geomec, you are able to visualize model elements and their properties.

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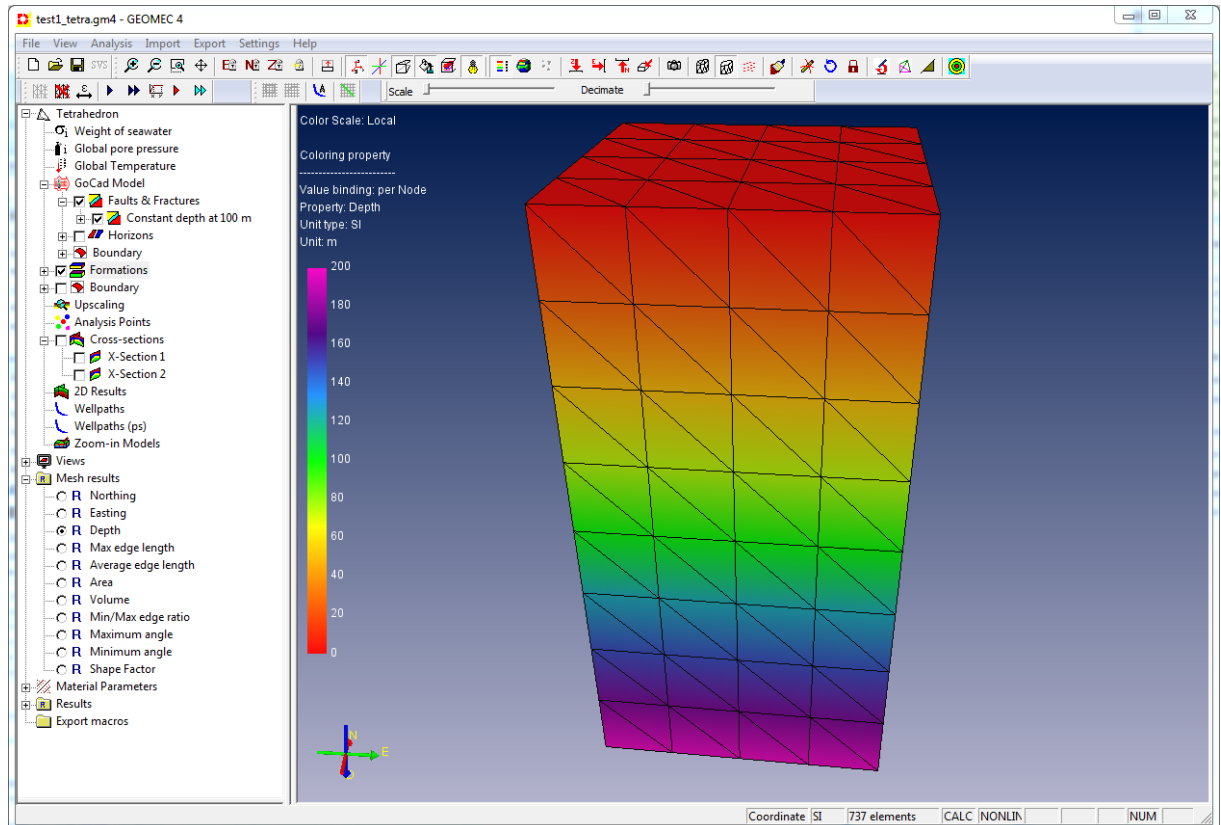


Figure 1: Open Inventor view with meshes and a property displayed.

As before you can toggle between a Value view and a Geology view using View toolbar menu item. On the left side of the Open Inventor window you find the legend and the sun dial.

3.2 Mouse behavior

With the left mouse button you can rotate the model displayed or manipulate the value tracker. Using the right mouse button you can drag the model around and using the scroll wheel you can pan (move the camera forward/backward). You can also use the Shift+ and the Shift- keyboard buttons for panning. Zooming is done using the zoom buttons



in the toolbar menu. Alternatively, you can use the Ctrl+, Ctrl- and Ctrl0 keyboard buttons.

3.3 The value/cross section dragger.

The value/cross section dragger allows you to track information on nodes and cells in the mesh or to manipulate the active cross section. The tracker will become visible if you

click the dragger-based value tracker button  or enable a cross section and it will look as shown in Figure 2. If the tracker hides inside your mesh you can make it visible

by toggling the fill mode button  or by increasing the transparency.

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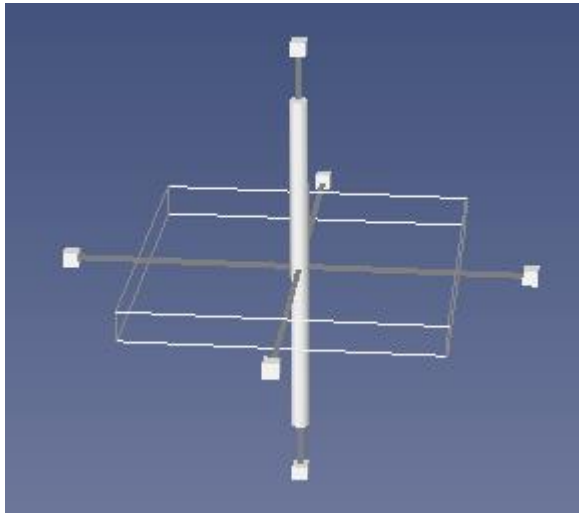


Figure 2: The dragger

The dragger, you can:

- resize by selecting one of the small cubes and dragging it.
- move axially by selecting the thick rod and dragging it along its axis.
- move laterally by selecting the flat square box and dragging it along its plane.
- rotate by selecting one of the thin rods and dragging it.

You may have to play around with this dragger a bit to get accustomed to its behavior. Apart from the dragger you may also see highlighted the mesh cell and mesh node that is currently being tracked. You can enable or disable this feature in the *View → Options...* dialog. Alternatively, you can click inside (not on the value tracker) or outside the model area to enable or disable this feature.

3.4 The mesh point information window.

In the bottom right corner of the screen you will find information regarding the point selected in the mesh, see Figure 3.

You can visualize this window by selecting either the dragger-based value tracker  or the mouse-based value tracker . If you select the mouse-based value tracker you can select a point for which to display information by left-clicking in the model area.

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```
Dragger orientation E: -0.348, N: 0.047, D: -0.936
Node Id: 138
Coordinates E: 50, N: 50, D: 100
Value: 50

Cell Id: 445
Coordinates E: 50.2894, N: 56.5208, D: 101.291
Value: 50.2894
```

Figure 3: Information on the point in the mesh selected by the value tracker.

3.5 Cross sections

Cross sections you can insert and select for display via the model tree view.

Inserting a cross section will pop up a dialog as show in Figure 4. This dialog will also pop up if you select the attributes entry in the context menu of a cross section.

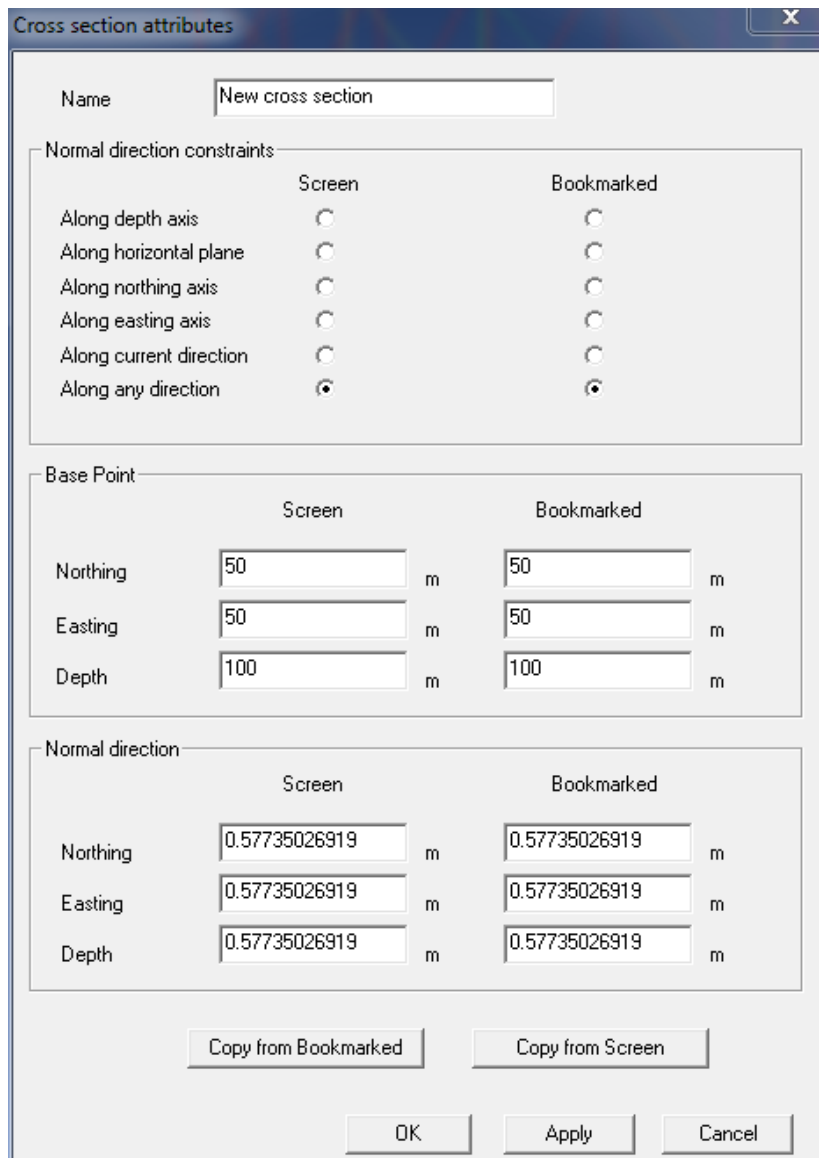
The right side of this dialog, called *Bookmarked*, contains the settings of a cross section as will be saved when you save your model. The left side, called *Screen*, contains the settings for a cross section as it is being displayed. The top part of this dialog contains settings to constrain the orientation and movement of the cross section.

You can copy the settings from *Bookmarked* to *Screen* and vice versa using the *Copy From* buttons. Click *OK* or *Apply* to save your settings.

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The dialog box is titled "Cross section attributes" and contains the following sections:

- Name:** A text field containing "New cross section".
- Normal direction constraints:** A table with two columns: "Screen" and "Bookmarked".

	Screen	Bookmarked
Along depth axis	☐	☐
Along horizontal plane	☐	☐
Along northing axis	☐	☐
Along easting axis	☐	☐
Along current direction	☐	☐
Along any direction	☒	☒
- Base Point:** A table with two columns: "Screen" and "Bookmarked".

	Screen	Bookmarked
Northing	50 m	50 m
Easting	50 m	50 m
Depth	100 m	100 m
- Normal direction:** A table with two columns: "Screen" and "Bookmarked".

	Screen	Bookmarked
Northing	0.57735026919 m	0.57735026919 m
Easting	0.57735026919 m	0.57735026919 m
Depth	0.57735026919 m	0.57735026919 m

At the bottom, there are two buttons: "Copy from Bookmarked" and "Copy from Screen". At the very bottom are three buttons: "OK", "Apply", and "Cancel".

Figure 4: Cross section attributes dialog

After you select a cross section for visualization it will pop up on the screen. Again, you may want to change transparency or toggle the Fill Mode to actually see the cross section. Together with the cross section you will see a dragger. This dragger you can use to move the cross section around.

In Geology mode, the cross section will show the colors of the formations in the model.

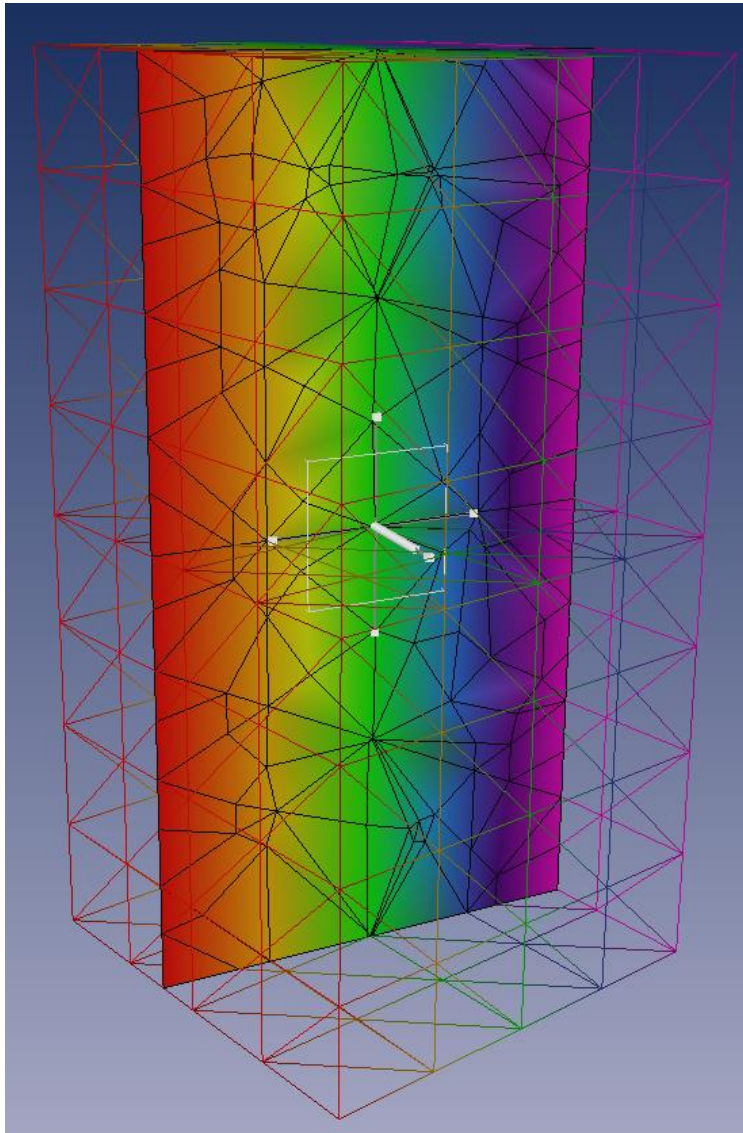


Figure 5: Cross section with value tracker.

Note: you can show multiple cross sections at the same time but only one (the last being visualized) will be under control of the value tracker.

By toggling the clipping button , you can clip away everything on one side of the cross section that is controlled by the value tracker.

3.6 Contour lines

You can visualize contour lines by selecting the Contour lines button .

If contour lines are visible you can change the property for which to display contour lines by selecting another property in the model tree view. See for example Figure 6.

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To make the number on the contour lines better readable, you can choose to toggle the fill mode or make the mesh more transparent using the transparency field in the *View → Options...* dialog. Alternatively, use the 'T' keyboard button with or without the Shift key to increase or decrease transparency. In that same dialog you can set the number of values for which to display contour lines.

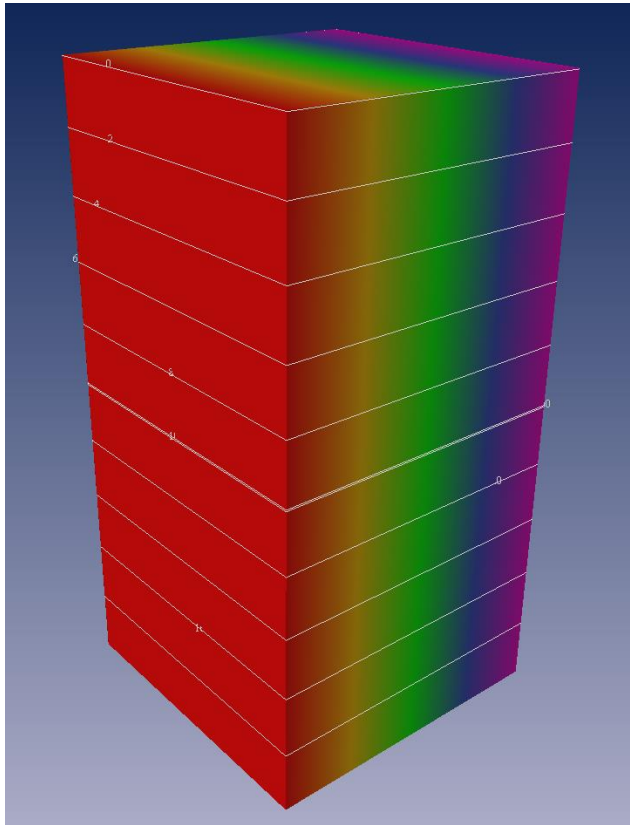


Figure 6: Display of contour lines for a property different from the coloring property.

3.7 Vectors

This Geomec version is able to display vector fields. In the display of vectors it distinguishes between regular vectors and vectors that are attributes of stress or strain tensors. A vector of regular vector field is displayed in white and contains an arrow to denote the direction of the vector. See for an example Figure 8.

The principal vectors of a tensor not only specify a size and direction, but through their eigenvalues, also denote whether the stress or strain associated is compressive or tensile. Principal vectors are expressed as colored lines that go through the vector's point of origin with their color denoting whether the stress/strain is compressive (green) or extensional (red).

Vectors are displayed if you select the *Full Vector* component of a vector. The displayed length and the number of vectors drawn can be manipulated via the *Scale* and *Decimate* slider, respectively. Note that in case of principal vectors, the mesh is colored according to the eigenvalue of the principal vector instead of (just) its length.

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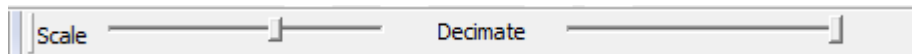


Figure 7: *Scale* and *Decimate* slider

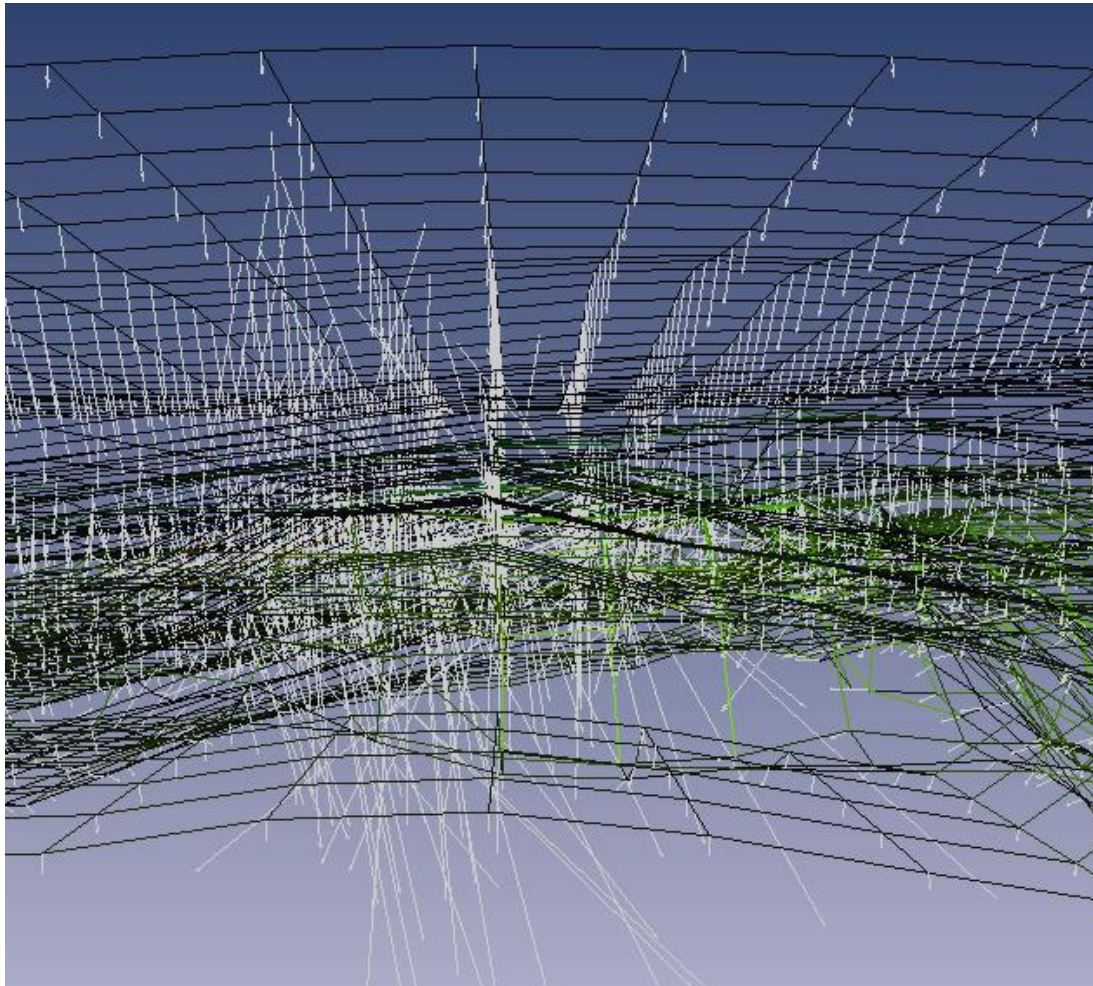


Figure 8: A number of surface meshes with a regular vector field.

4 Tensor display

Tensor properties can be displayed on a mesh as "beach balls" by selecting the *Full Tensor* component of a tensor property, e.g. stress. The mesh itself is colored according to the Frobenius norm of the tensor property.

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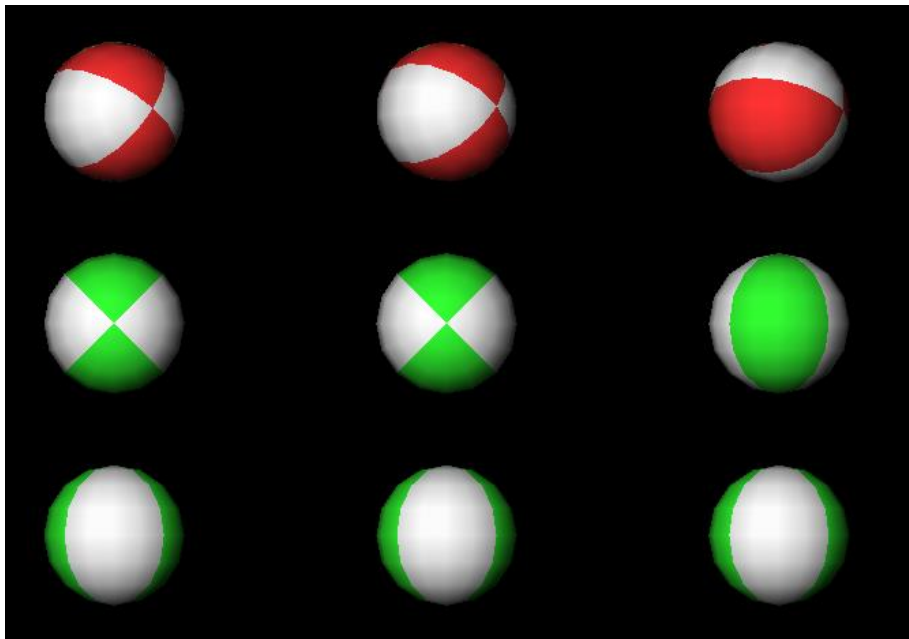


Figure 9: Some examples of tensor beach balls.

Beach balls are used to display the three principal directions of a tensor in a single picture.

The principal direction with the greatest eigenvalue is oriented along the points where the white areas and the red/green areas of the beach ball meet.

The principal direction with the smaller eigenvalue is oriented along the middle points of the two red/green areas. If the smallest eigenvalue is positive, the color is green, else it is red.

Conversely, the principal direction with the intermediate eigenvalue is oriented along the middle points of the two white areas.

Geomec also contains functionality to display the result of a tensor property as it is applied to a cross section. You can select a tensor property by selecting one of its components (NN, EE, DD, etc.) and display its effect on a displayed cross section. With

the buttons , you can select which vector component (full, normal or shear) to display.

5 Keyboard short cuts

M	Toggle the display of Mesh lines.
O	Toggle the display of formation outlines on cross sections.
T/Shift T	Decrease /Increase the transparency of the model with 10 %.
Ctrl+/Ctrl-	Zoom in or Zoom out.
Ctrl0	Reset zoom and camera position.
Shift+/Shift-	Move the camera into or out of the scene.
Escape	Abort an active rubber banding operation.
Return	Finalize an active rubber banding operation.

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6 2D views

The history and well path results views have been redesigned to fit user expectations better.

- Both views can display multiple curves, e.g. allowing for multiple well paths to be displayed together or for multiple depletion stages or properties on a single well.
- Both views also allow for curves with different units or different ranges to be displayed simultaneously using different axes to display their ranges. Right-click in the view to change your preferred axis for subsequent curve assignment. If a property cannot be assigned to the preferred axis because of a unit mismatch, it will be assigned to the other axis.
- Dotted curves relate to the bottom (well view) or right axis (history view), solid curves relate to the top (well view) or left axis (history view).
- Both views allow to set foreground and background colors and to set curve line thickness, e.g. to create snapshots for inclusion in documents. See the *View → Options...* dialog.
- The well path results view allows for selection of either the AHD axis or the TVDKB axis to be the linear vertical axis with the other axis plotted alongside (right-click in the view). In case of multiple wells being displayed the depth with respect to mean sea level will be displayed on the vertical axis.
- If you do not like the color of a curve, for instance because it is similar to the color of another curve, you can easily change its color by deselecting and selecting the check box related to the curve.

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Well Path: GK-W502B
Unit type: SI
Depletion stage: Pre-production 2012

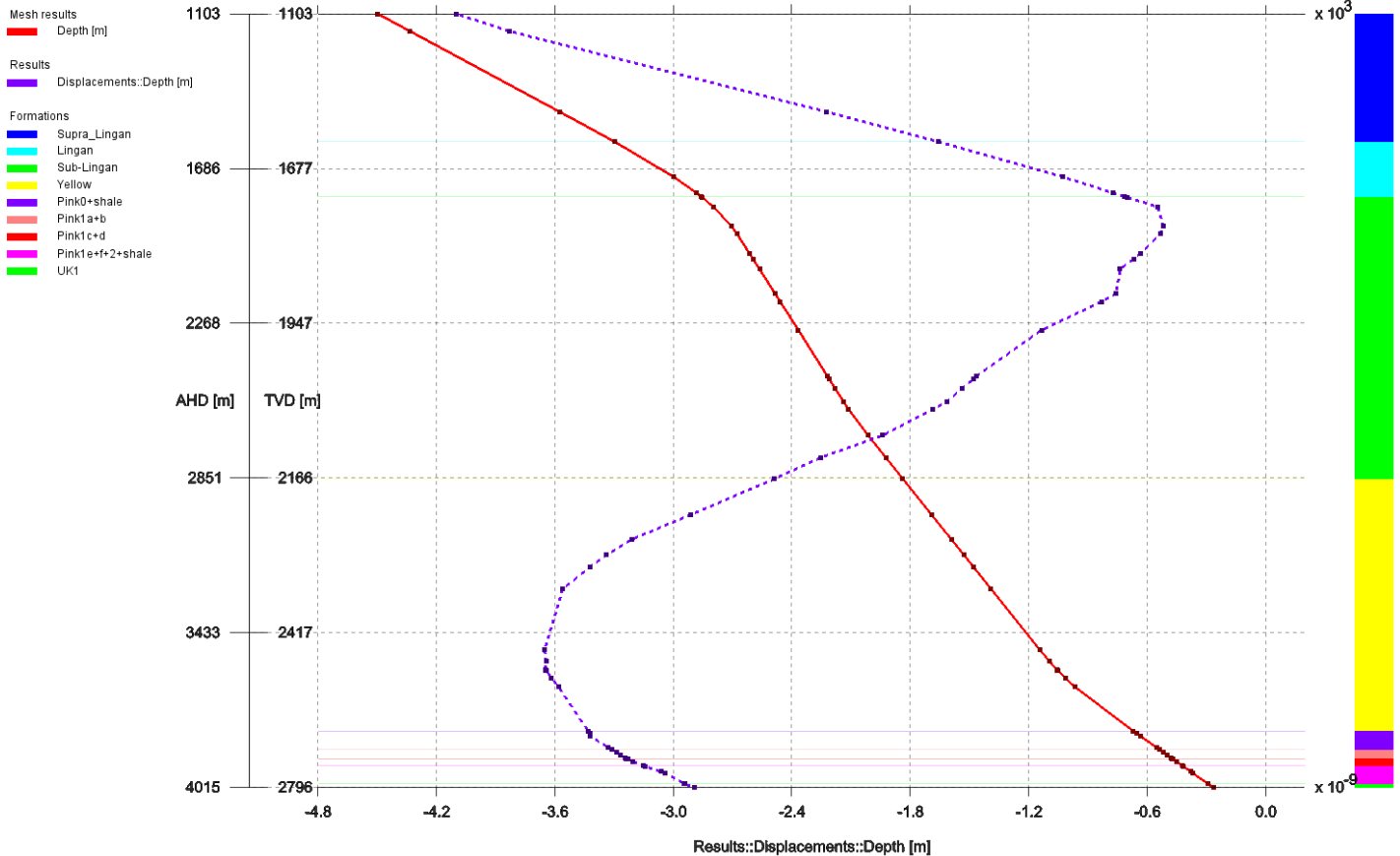


Figure 10: Example well path results view with a single well and two horizontal axes and with the AHD axis as the linear vertical axis. Dotted curves relate to the bottom axis, solid curves relate to the top axis.



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Analysis: Non-linear
Unit type: SI

Results::Displacements::Easting [m]

Analysis Points:
New point0
New point1
New point2

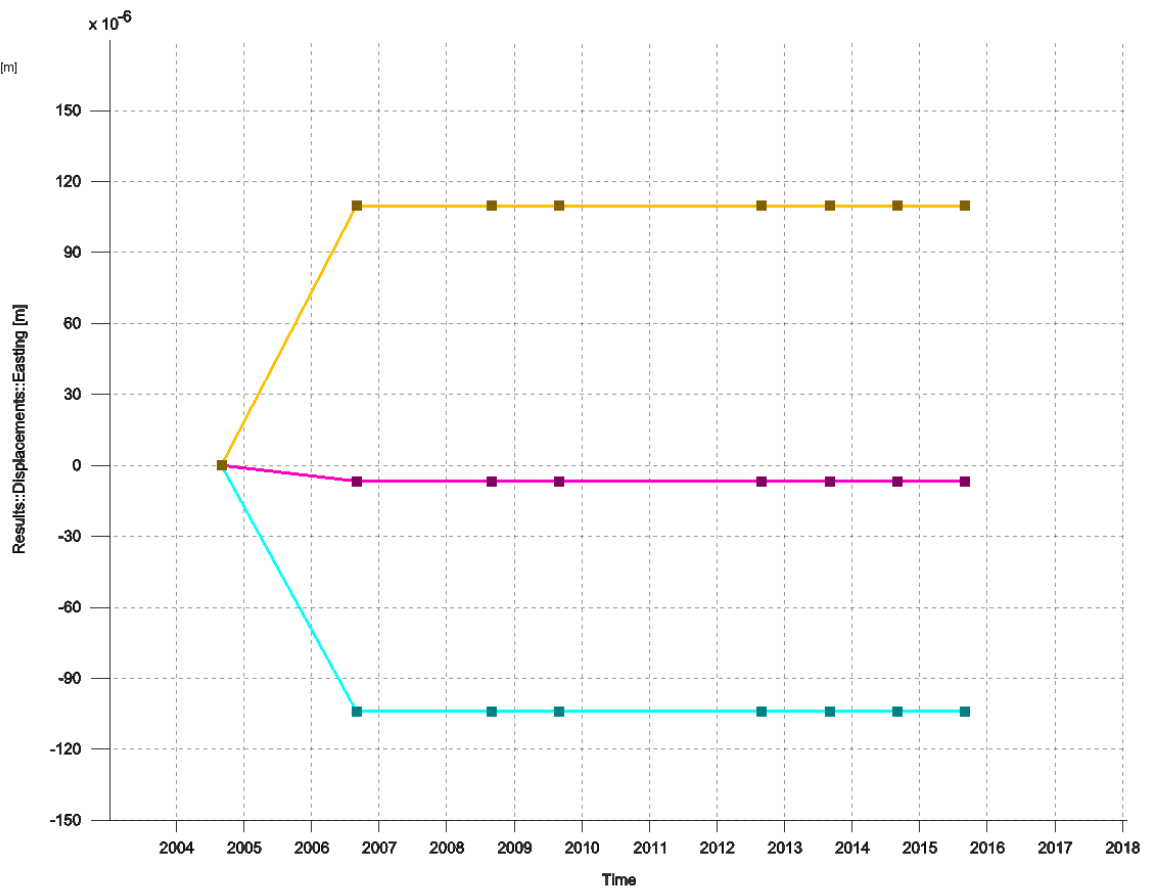


Figure 11: Example history view.